Question	Which of the following lack cells?
Type	multiple choice
Option	Humans
Option	Plants
Option	Bacteria
Option	Viruses
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	DPLP size is greater then-
Type	multiple choice
Option	Ribosome
Option	Mitochondria and plastids
Option	RBC
Option	Both (b) and (c)
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Mark the correctly matched -
Type	multiple choice
Option	RBC - biconcave and elongated
Option	Mesophyll cell - elongated and branched
Option	Columnar epithelium - round and spherical
Option	Tracheids – Elongated
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	In prokaryotes, structure which is not part of cell envelope
Type	multiple choice
Option	Cell wall
Option	Cell membrane
Option	Flagella
Option	Glycocalyx
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Match the following:

Column – I		Column – II	
(i)	Robert brown	(a)	Dead cork cell
(ii)	Robert hooke	(b)	Nucleus
(iii)	Leeuwenhoek	(c)	Triple helical structure of collagen
(iv)	G.N Ramachandr an	(d)	Living cell

Type	multiple choice

Option	i-b, ii-a, iii-d, iv-c
Option	i-a, ii-b, iii-c, iv-d
Option	I-b, ii-c, iii-a, iv-d
Option	i-b, ii-a, iii-c, iv-d
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following statement is true for a bacteria?
Type	multiple choice
Option	Spirillum is shape present for some bacteria
Option	Cytoplasm is place for all cellular activities
Option	The cells of the human cheek have an outer membrane as the delimiting structure of the cell
Option	Animal cells contain another membrane bound organelle called centrosome which helps in cell division
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	What is external to the cell wall in a prokaryotic cell?
Type	multiple choice
Option	Glycocalyx layer
Option	Plasma membrane
Option	Both (a) and (b)
Option	None of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following statement is wrong?
Type	multiple choice
Option	The ribosomes of a polysome translate the RNA into proteins
Option	Glycocalyx differs in composition and thickness among different bacteria.
Option	Most prokaryotic cells, particularly the bacterial cells, have a chemically complex cell envelope.
Option	Smaller DNA is known as genomic DNA. The genomic DNA confers certain unique phenotypic characters to such bacteria
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which statement is correct?
Type	multiple choice
Option	A specialised differentiated form of cell membrane called mesosome is the characteristic of prokaryotes.
Option	Cell envelope consists of a tightly bound four layered structure.
Option	Glycocalyx could be a loose sheath called the slime layer in some, while in others it may be thick and tough, called the capsule.
Option	The prokaryotic cells are represented by bacteria, blue-green algae, mycoplasma and PPLO.
Answer	2
Solution	
Positive Marks	4

Negative Marks	1
Question	Which of the following is **not** role of mesosomes
Type	multiple choice
Option	Respiration
Option	Photosynthesis
Option	Secretion
Option	Cell wall formation
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Mark the **incorrect** matched pair-
Type	multiple choice
Option	Mesosome - Vesicle and tubules shape
Option	Chromatophore - Extensions of membrane
Option	Fimbriae - Bristle shape
Option	Pilli - Filamentous and for attachment
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Chromatophore present in
Type	multiple choice
Option	Mycoplasma
Option	Streptococcus
Option	Lactobacillus
Option	Nostoc
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Structure absent in Nostoc
Type	multiple choice
Option	Cell wall
Option	Cell membrane
Option	Gas vacuole
Option	Flagella
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Structure not involve any membrane
Type	multiple choice
Option	Ribosome
Option	Chromatophore
Option	Nucleus
Option	ER
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Polyribosomes are _____.
Type	multiple choice

Option	Single ribosome
Option	Many ribosomes on mRNA
Option	Many ribosomes on tRNA
Option	Inclusion
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Cell wall is absent in
Type	multiple choice
Option	Cyanobacteria
Option	Archaebacteria
Option	Fungi
Option	Amoeba
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Bacteria show a range in the number of arrangement of flagella. Bacterial flagellum is composed of
Type	multiple choice
Option	Two parts - pili and fimbriae
Option	Three parts - filament, hook and basement membrane
Option	Three parts - filament, shaft and basal body
Option	Three parts - filament, hook and basal body
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Structure which is present in all eukaryotic cell
Type	multiple choice
Option	Plasma membrane
Option	Mesosomes
Option	Cell wall
Option	All
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Similarity present in plasmid and nucleoid
Type	multiple choice
Option	Present in eukaryotic cell
Option	Single DNA
Option	Present in all prokaryotic cell
Option	Circular DNA
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Cell membranes possess lipid, protein and carbohydrate. The ratio of protein and lipid varies considerably in different cell types. In human beings, the membrane of the RBCs has approximately
Type	multiple choice

Option	40 percent lipids and 52 percent carbohydrates
Option	40 percent protein and 52 percent lipids
Option	40 percent lipids and 52 percent proteins
Option	40 percent lipids and 52 percent carbohydrate
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following promotes the movement of molecules when carrier-mediated facilitated diffusion is involved?
Type	multiple choice
Option	Concentration gradient
Option	Availability of carrier molecules
Option	Temperature
Option	Both (a) and (b)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	What is true regarding fluid mosaic model?
Type	multiple choice
Option	Phospholipid monolayer is present over protein layer
Option	Phospholipid bilayer is present over protein layer
Option	Protein embedded in phospholipid bilayer
Option	Phospholipid layer is sandwiched between two protein layers
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Molecules which are transported across the membrane against their concentration gradient, i.e., from the lower to higher concentration. Such a transport is called
Type	multiple choice
Option	Active transport, e.g., diffusion
Option	Passive transport, e.g., diffusion
Option	Active transport, e.g., Na+/K+ pump
Option	Osmosis, a type of simple diffusion
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following layer is present nearest plasma membrane in plant cell?
Type	multiple choice
Option	Tonoplast
Option	Middle lamella
Option	Primary wall
Option	Secondary wall
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Cell wall is made up of
Type	multiple choice

Option	Several layers of microfibrils
Option	Synchronous mitotic division
Option	Single layer of Cellulose molecules
Option	Glucose molecules
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Function(s) of the cell wall is/are
Type	multiple choice
Option	Provide shape of the cell and protects the cell from the mechanical damage and infection
Option	Helps in cell-to-cell interaction
Option	Provides barrier to undesirable macromolecules
Option	All of the above
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which is **not** modification of cell membrane
Type	multiple choice
Option	Mesosomes
Option	Chromatophore
Option	Microvillus
Option	Inclusion body
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	**S-I:** Cell wall is present in Monera, Fungi and Plantae **S-II:** Monera and Plantae both have cell wall of cellulose
Type	multiple choice
Option	Only S-I is correct
Option	Only S-II is correct
Option	Both S-I and S-II correct
Option	Both S-I and S-II wrong
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Common in algae and plant cell wall is
Type	multiple choice
Option	Cellulose
Option	Hemicellulose
Option	Pectin
Option	All of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which is **not** a feature of cell wall
Type	multiple choice
Option	Rigid

Option	Provide structure support
Option	Semi-permeable
Option	Protect from infection
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the face, Golgi complex is associated with ER?
Type	multiple choice
Option	Forming face, i.e., Trans-face
Option	Maturing face, i.e., Trans-face
Option	Both forming and maturing face
Option	Forming face or Cis-face
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Organelle involved in modification and routing of newly synthesised proteins to their destination is
Type	multiple choice
Option	Chloroplast
Option	Lysosome
Option	Mitochondria
Option	Golgi Apparatus
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which is a part of endomembrane system of eukaryotic cells?
Type	multiple choice
Option	Mitochondria
Option	Peroxisome
Option	Chloroplasts
Option	Golgi bodies
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Hydrolytic enzymes are abundantly found in which cell organelles?
Type	multiple choice
Option	Ribosome
Option	Lysosome
Option	Oxysome
Option	Endoplasmic reticulum
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following statement is true for a secretory cell?
Type	multiple choice
Option	Golgi apparatus is absent
Option	Rough Endoplasmic Reticulum (RER) is easily observed in the cell.
Option	Only Smooth Endoplasmic Reticulum (SER) is present

Option	Secretory granules are formed in nucleus
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Membrane covering the vacuole is termed as
Type	multiple choice
Option	Cell wall
Option	Plasmalemma
Option	Cell membrane
Option	Tonoplast
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Similarity between ER and Golgi is
Type	multiple choice
Option	Present in eukaryotes
Option	Part of endomembrane system
Option	Made up of lipid bilayer
Option	All
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	80S type of ribosomes are found in
Type	multiple choice
Option	Prokaryotic cells
Option	Chloroplasts
Option	Mitochondria
Option	Cytoplasm of eukaryotic cell
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Sedimentation coefficient(s) is indirectly a measure of
Type	multiple choice
Option	Density
Option	Size
Option	Volume
Option	Both (a) and (b)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Oxidative phosphorylation in a cell occurs in
Type	multiple choice
Option	Cell wall
Option	Ribosomes
Option	Mitochondria
Option	Plastids
Answer	3
Solution	

Positive Marks	4
Negative Marks	1
Question	Which of the following is **not true** about mitochondria?
Type	multiple choice
Option	Double stranded DNA
Option	70S ribosome
Option	Divided by fission
Option	Use CO_2
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The plastids that are involved in storage of fat are
Type	multiple choice
Option	Aleuroplast
Option	Amyloplasts
Option	Elaioplasts
Option	Oleosomes
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Chloroplast does not contain
Type	multiple choice
Option	Double membraned envelope
Option	Grana and Stroma
Option	Matrix and Cristae
Option	Circular DNA and 70S ribosomes
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is correctly match?
Type	multiple choice
Option	Mitochondria - Food production
Option	Ribosomes - Protein synthesis
Option	Lysosomes - Kitchen of cell
Option	Aleuroplast - Store Oils and fats
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following statements regarding mitochondria is **incorrect**?
Type	multiple choice
Option	Mitochondria are the sites of aerobic respiration.
Option	The outer compartment is called the matrix.
Option	The inner membrane forms a number of infoldings called the cristae towards the matrix.
Option	Mitochondrial matrix contains single circular DNA molecule and ribosomes.
Answer	2
Solution	
Positive Marks	4
Negative Marks	1